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ELECTRONIC SYSTEM INCLUDING CENTRAL CONTROLLER AND ZONAL CONTROLLER AND VEHICLE INCLUDING THE SAME

Abstract

An electronic system may include a central controller configured to, receive speed information related to a current speed of a vehicle, generate target speed information representing a speed at which the vehicle will travel based on the speed information, and transmit the target speed information, and a first zonal controller configured to control a first zone among a plurality of zones of the vehicle, and a second zonal controller configured to control a second zone among the plurality of zones of the vehicle, and control at least one motor of the vehicle based on the target speed information received from the central controller, the at least one motor configured to drive at least one wheel of the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This U.S. non-provisional application claims the benefit of priority to Korean Patent Application No. 10-2024-0028262 filed in the Korean Intellectual Property Office on Feb. 27, 2024, the entire contents of which is incorporated herein by reference.

BACKGROUND

[0002] Various example embodiments of the inventive concepts relate to an electronic system including a central controller and a zonal controller, a vehicle including the same, and/or methods for operating the electronic system, etc.

[0003] The electronic system of a conventional vehicle is a system in which the controllers that control the powertrain, chassis, body, in-vehicle infotainment (IVI), and/or advanced driver assistance system (ADAS) are distributed. Currently, the electronic system of a vehicle is changing to a centralized electronic system in which the multiple zones of the vehicle are controlled by multiple zone controllers, and the multiple zone controllers are controlled by a central controller capable of high-performance computing.

SUMMARY

[0004] Various example embodiments of the inventive concepts provide an electronic system including a central controller and a zonal controller, a vehicle including the same, and/or a method of operating the electronic system, etc., capable improving performance for controlling the vehicle in the longitudinal direction.

[0005] An electronic system may include a central controller configured to, receive speed information related to a current speed of a vehicle, generate a target speed information representing a speed at which the vehicle will travel based on the speed information, and transmit the target speed information, and a first zonal controller configured to control a first zone among a plurality of zones of the vehicle, and a second zonal controller configured to control a second zone among a plurality of zones of the vehicle, and control at least one motor of the vehicle based on the target speed information received from the central controller, the at least one motor configured to drive at least one wheel of the vehicle.

[0006] An electronic system may include an inverter configured to convert a voltage of a battery to a driving voltage, at least one motor configured to operate based on the driving voltage, a plurality of zonal controllers configured to control each of a plurality of zones of the vehicle, and a central controller configured to determine target speed information based on current rotation speed information of the at least one motor and desired speed information of a vehicle housing the at least one motor, the target speed information indicating a target speed at which the vehicle will travel, and the desired speed information indicating a desired speed to which the vehicle will increase or decrease. Wherein one zonal controller for controlling one of the plurality of zones of the vehicle is configured to receive the target speed, and adjust the driving voltage based on a target current corresponding to the target speed information and a driving current corresponding to the driving voltage.

[0007] A vehicle may include a first motor and a second motor each configured to drive at least one first wheel and at least one second wheel, respectively, a central controller configured to generate target speed information and speed change information of the vehicle, the target speed information representing a speed at which the vehicle will travel based on rotation speed information of the first motor and the second motor, a first zonal controller configured to control a first zone among a plurality of zones of the vehicle, and control a first driving voltage applied to the first motor based on the target speed information, and a second zonal controller configured to control a second zone among the plurality of zones of the vehicle, and control a second driving voltage applied to the second motor based on the target speed information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a drawing illustrating an example vehicle including an electronic system according to at least one example embodiment.

[0009] FIG. **2** is a block diagram of an example electronic system including a central controller and a zonal controller according to at least one example embodiment.

[0010] FIG. **3** is a block diagram of an example central controller according to at least one example embodiment.

[0011] FIG. **4** is a block diagram of an example zonal controller according to at least one example embodiment.

[0012] FIG. **5** is a diagram of an example inverter, motor, and zonal controller according to at least one example embodiment.

[0013] FIG. **6** is a block diagram of example zonal controllers for controlling a plurality of motors according to at least one example embodiment.

[0014] FIG. **7** is a flowchart illustrating an example method of operating an electronic system according to at least one example embodiment.

[0015] FIG. **8** is a flowchart illustrating an example method of operating a zonal controller according to at least one example embodiment.

DETAILED DESCRIPTION

[0016] Various example embodiments of the inventive concepts will be described more fully hereinafter with reference to the accompanying drawings. As those of ordinary skill in the art would realize, the described example embodiments may be modified in various different ways, all without departing from the spirit or scope of the inventive concepts.

[0017] The drawings and description are to be regarded as illustrative in nature and not restrictive, and like reference numerals designate like elements throughout the specification.

[0018] In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of stated elements but not the exclusion of any other elements.

[0019] FIG. **1** is a drawing illustrating an example vehicle including an electronic system according to at least one example embodiment.

[0020] Referring to FIG. **1**, a vehicle **10** may include an electronic system **50**, but is not limited thereto. The electronic system **50** may be a system for processing data generated by the vehicle **10**. In at least one example embodiment, the vehicle **10** may include a plurality of zones, e.g., a first zone Z**1**, a second zone Z**2**, a third zone Z**3**, and/or a fourth zone Z**4**, etc., but is not limited thereto, and for example may include a greater or lesser number of zones, etc.

[0021] The electronic system **50** may include a plurality of zonal controllers, e.g., a first zonal controller **100**, a second zonal controller **200**, a third zonal controller **300**, a fourth zonal controller **400**, etc., a central controller **500**, a sensor **600**, a battery **403**, an inverter **404**, a motor **405**, a resolver **406**, and/or a plurality of micro control units (MCU) **101**, **102**, **201**, **202**, **301**, **302**, **401**, and **402**, etc., but the example embodiments are not limited thereto. According to some example embodiments, one or more of the first zonal controller **100**, the second zonal controller **200**, a third zonal controller **300**, a fourth zonal controller **400**, the central controller **500**, the sensor **600**, and/or a plurality of micro control units (MCU) **101**, **102**, **201**, **202**, **301**, **302**, **401**, **402**, etc., may be implemented as processing circuitry. Processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0022] The plurality of micro control units (MCU), e.g., MCUs **101**, **102**, **201**, **202**, **301**, **302**, **401**, and/or **402**, etc., may control operations performed in and/or by the vehicle **10** based on information obtained from the inside and/or the outside of the vehicle. For example, one or more of the plurality of micro control units (MCU) **101**, **102**, **201**, **202**, **301**, **302**, **401**, and **402** may control the heating wire of the vehicle seat based on temperature information received from a temperature sensor within the vehicle **10**, etc., but the example embodiments are not limited thereto, and the plurality of MCUs may control other features of the vehicle **10**, such as steering, accelerating, braking, air conditioning, heating, displays, lighting, communication, entertainment, charging, etc.

[0023] The first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and/or the fourth zonal controller **400** may control one or more of the plurality of micro control units (MCU) **101**, **102**, **201**, **202**, **301**, **302**, **401**, and/or **402**, etc.

[0024] In at least one example embodiment, the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and the fourth zonal controller **400** may control the operations performed by vehicle equipment associated with and/or corresponding to the first zone Z**1**, the second zone Z**2**, the third zone Z**3**, and the fourth zone Z**4**, respectively, but are not limited thereto. According to some example embodiments, the zonal controllers may be dedicated controllers for performing a subset of tasks and/or managing a subset of equipment of the vehicle **10**, and accordingly, the zonal controllers may have less processing power, less memory resources, etc., than the central controller **500** thereby reducing the costs of the electronic system **50**, but the example embodiments are not limited thereto, and for example the zonal controllers may have the same and/or greater processing power and/or memory resources than the central controller **500**, etc. Further, because the control of individual sub-systems of the vehicle **10** may be performed by multiple zonal controllers based on inputs from the central controller **500**, the transmission speed and/or latency of transmitting instructions and/or commands may be reduced, and the probability of instructions and/or commands not being received by vehicle equipment and/or sub-systems is reduced in comparison to having the central controller **500** control all of the vehicle equipment, thereby improving the functionality and/or safety of the vehicle **10**.

[0025] In at least one example embodiment, the first zonal controller **100** may control operation performed in the first zone Z**1**. For example, the first zone Z**1** may be an area corresponding to front left side of the vehicle **10**, but is not limited thereto, and for example the first zone Z**1** may refer to at least one component and/or sub-system of the vehicle **10**, such as the heating and air conditioning (HVAC) subsystem of the vehicle **10**, the driving subsystem of the vehicle **10**, the entertainment subsystem of the vehicle **10**, etc. The first zonal controller **100** may receive data from MCUs **101** and **102**, and may control operation of the MCUs **101** and **102**, etc. For example, the first zonal controller **100** may control the MCUs **101** and **102** to increase or decrease the temperature of a front left seat based on temperature information received from the temperature sensor within the vehicle **10**, etc.

[0026] In at least one example embodiment, the second zonal controller **200** may control operation performed in the second zone Z**2**. For example, the second zone Z**2** may be an area corresponding to front right side of the vehicle **10**, but is not limited thereto, and for example the second zone Z**2** may refer to at least one component and/or sub-system of the vehicle **10**, etc. The second zonal controller **200** may receive data from MCUs **201** and **202**, and may control operation of the MCUs **201** and **202**, etc. For example, the second zonal controller **200** may control the MCUs **201** and **202** to increase or decrease the temperature of a front right seat based on temperature information received from the temperature sensor within the vehicle **10**, etc.

[0027] In at least one example embodiment, the third zonal controller **300** may control operation performed in the third zone Z**3**. For example, the third zone Z**3** may be an area corresponding to rear left side of the vehicle **10**, but is not limited thereto, and for example the third zone Z**3** may refer to at least one component and/or sub-system of the vehicle **10**, etc. The third zonal controller **300** may receive data from MCUs **301** and **302**, and may control operation of the MCUs **301** and **302**, etc. For example, the third zonal controller **300** may control the MCUs **301** and **302** to increase or decrease the temperature of a rear left seat based on temperature information received from the temperature sensor within the vehicle **10**, etc.

[0028] In at least one example embodiment, the fourth zonal controller **400** may control operation performed in the fourth zone Z**4**. For example, the fourth zone Z**4** may be an area corresponding to rear right side of the vehicle **10**, but is not limited thereto, and for example the fourth zone Z**4** may refer to at least one component and/or sub-system of the vehicle **10**, etc. The fourth zonal controller **400** may receive data from MCUs **401** and **402**, and may control operation of the MCUs **401** and **402**, etc. For example, the fourth zonal controller **400** may control the MCUs **401** and **402** to increase or decrease the temperature of a rear right seat based on temperature information received from the temperature sensor within the vehicle **10**, etc. However, the example embodiments are not limited to the at least one example embodiment illustrated in FIG. **1**, and for example, the vehicle **10** may include a greater or lesser number of zones, a greater or lesser number of zonal controllers, and/or a greater or lesser number of MCUs, etc.

[0029] The battery **403** may store electrical energy used to operate and/or drive the vehicle **10**. In at least one example embodiment, the battery **403** may supply a direct current (DC) power.

[0030] The inverter **404** may convert the voltage of the battery **403** to a driving voltage, and may apply the driving voltage to the motor **405**. In at least one example embodiment, the inverter **404** may convert DC power of the battery **403** to the driving voltage corresponding to alternating current (AC) power.

[0031] According to at least one example embodiment, the at least one motor **405** may be an electric motor for a battery electric vehicle (BEV) and/or a hybrid electric vehicle (e.g., a vehicle with an electric motor and an internal combustion engine) and may operate and/or drive the vehicle **10** based on the driving voltage applied and/or received from the inverter **404**, but the example embodiments are not limited thereto, and for example, the motor **405** may be an internal combustion engine for a gasoline and/or diesel powered vehicle, a hybrid electric vehicle, etc. In the example embodiments where the motor **405** is an internal combustion engine, the driving voltage may be a control signal for a throttle control system (e.g., drive-by-wire throttle system, etc.) for the internal combustion engine, etc., but is not limited thereto. In at least one example embodiment, the motor **405** may perform rotational movement according to and/or based on the driving voltage, and the vehicle **10** may be driven as at least one wheel connected to the motor **405** is rotated by the motor **405**. In at least one example embodiment, the at least one wheel connected to the motor **405** may be located in the rear of the vehicle. In at least one other example embodiment, the at least one wheel connected to the motor **405** may be located in the front of the vehicle. In some example embodiments, the motor **405** may be a plurality of motors each connected to one or wheels of the vehicle **10**, but the example embodiments are not limited thereto.

[0032] The resolver **406** may detect a rotor position of a rotor and/or shaft rotating in the motor **405**. The resolver **406** may provide information on the rotor position (e.g., an angle and/or displacement speed, etc.) to a zonal controller controlling the motor **405**, but is not limited thereto. Additionally, or alternatively, an encoder, etc., may be used to detect the rotor position information of the motor **405**.

[0033] The central controller **500** may control overall operation of the electronic system **50**. In at least one example embodiment, the central controller **500** may control the operation of the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and/or the fourth zonal controller **400**, etc. In at least one example embodiment, the central controller **500** may communicate data with the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and/or the fourth zonal controller **400**, etc., through at least one communications bus, such as an Ethernet network bus, but the example embodiments are not limited thereto, and for example, other communication protocols and/or interfaces may be implemented.

[0034] In at least one example embodiment, the central controller **500** may process data related to an advanced driver assistance system (ADAS), but is not limited thereto. In at least one example embodiment, ADAS may include one or more functions, such as lane departure warning (LDW), lane keeping assist (LKA), high beam assist (HBA), autonomous emergency braking (AEB), traffic sign recognition (TSR), smart cruise control (SCC), blind spot detection (BSD), and/or forward collision-avoidance assist (FCA), etc., but is not limited thereto.

[0035] In at least one example embodiment, the central controller **500** may process data related to in-vehicle infotainment (IVI), etc. For example, the central controller **500** may control audio and/or video information related to vehicle settings (e.g., heating and/or cooling settings, seat settings, lighting settings, driving settings, etc.), navigation functions, communication functions (e.g., phone calls, emails, text messages, etc.), entertainment (e.g., music, movies, TV, games, etc.), etc., to be output through at least one display in the vehicle **10**.

[0036] In at least one example embodiment, the central controller **500** may receive speed information related to a speed of the vehicle **10**, and may generate target speed information representing a desired speed at which the vehicle **10** will travel based on the speed information, etc. In at least one example embodiment, the speed information may include travel speed information representing a current speed of the traveling vehicle **10**, speed change information representing a speed to which the vehicle **10** will increase or decrease, gear information representing a gear position of the transmission, and/or rotation speed information representing a speed at which the motor **405** rotates, etc., but is not limited thereto.

[0037] In at least one example embodiment, the central controller **500** may receive the travel speed information, the speed change information, and/or the gear information, etc., from the at least one sensor **600**. In at least one example embodiment, the central controller **500** may receive the rotation speed information from the zonal controller controlling the motor **405** among first to fourth zonal controllers **100**, **200**, **300**, and/or **400**, etc. In at least one example embodiment, the central controller **500** may provide the target speed information to the zonal controller controlling the motor **405**, etc.

[0038] The at least one sensor **600** may detect the current speed of the traveling vehicle, a level of depressing the pedal(s) by a driver driving and/or operating the vehicle, the current revolutions per minute (RPMs) of the motor **405**, and/or the position of the transmission, etc. The sensor **600** may generate the travel speed information based on the current speed of the traveling vehicle, etc. The sensor **600** may generate the speed change information based on the level of depressing the pedal(s) by the driver. The sensor **600** may generate the gear information based on the position of the transmission. The sensor **600** may provide the travel speed information, the speed change information, and/or the gear information, etc., to the central controller **500**. In at least one example embodiment, the sensor **600** may be located in the first zone Z**1** where the driver is seated, but is not limited thereto. For example, the sensor **600** may include an accelerator pedal sensor, a brake pedal sensor, a clutch pedal sensor, a transmission selection sensor, or the like, that may be manipulated by the driver. The sensor **600** may be connected to the first zonal controller **100**, etc. The information pieces generated from the sensor **600** may be provided to the central controller **500** through the first zonal controller **100**, but is not limited thereto.

[0039] In at least one example embodiment, one zonal controller among the first zonal controller **100**, the second zonal controller **200**, the third zonal controller **300**, and the fourth zonal controller **400**, etc., may control the motor **405** configured to drive the vehicle **10**, but the example

embodiments are not limited thereto. In FIG. **1**, it is assumed, for the sake of clarity and brevity, that the fourth zonal controller **400** controls the motor **405**, but the example embodiments are not limited thereto.
[0040] In at least one example embodiment, the fourth zonal controller **400** may receive the target speed information from the central controller **500**. The fourth zonal controller **400** may control the motor **405** configured to drive the vehicle **10** based on the target speed information. In at least one example embodiment, the fourth zonal controller **400** may generate a pulse width modulation (PWM) signal that controls a width of the driving voltage to be applied to the motor **405** based on the target speed information. In at least one example embodiment, the fourth zonal controller **400** may control the inverter **404** to turn-on and/or turn-off switches included in the inverter **404** that apply the driving voltage according to and/or based on the PWM signal, etc.
[0041] FIG. **2** is a block diagram of an example electronic system including a central controller and a zonal controller according to at least one example embodiment.
[0042] Referring to FIG. **2**, the electronic system **50** may include the fourth zonal controller **400**, the battery **403**, the inverter **404**, the motor **405**, the resolver **406**, the central controller **500**, and/or the sensor **600**, etc., but the example embodiments are not limited thereto, and for example, the electronic system **50** may include a greater or lesser number of constituent components.
[0043] In at least one example embodiment, the central controller **500** may control the fourth zonal controller **400**, but is not limited thereto. The central controller **500** may include a central processor **510**, a central memory **520**, a communication interface (e.g., a network interface, an Ethernet interface, etc.) **530**, a sensor interface **540**, and/or a storage device **550**, etc. According to some example embodiments, one or more of the central processor **510**, the communication interface **530**, the sensor interface **540**, etc., may be implemented as processing circuitry. Processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.
[0044] The central processor **510** may control the overall operation of the central controller **500**. In at least one example embodiment, the central processor **510** may execute a vehicle control module **560** stored in the central memory **520**, etc. In at least one example embodiment, the central processor **510** may determine a target speed corresponding to the speed at which the vehicle **10** will travel based on the speed information related to the speed of the vehicle received from the sensor **600** and/or the fourth zonal controller **400**, etc.
[0045] The central memory **520** may be used as a buffer memory, a cache memory, an operating memory, or the like of the central controller **500**. In at least one example embodiment, the central memory **520** may store the vehicle control module **560**, etc. In at least one example embodiment, the central memory **520** may be a volatile memory device and/or non-volatile memory device. For example, the central memory **520** may be a random-access memory (RAM) and/or read-only memory (ROM), etc.
[0046] In at least one example embodiment, when the central memory **520** is a volatile memory device, the vehicle control module **560** may be stored in a non-volatile memory device **552** of the storage device **550**, and the central memory **520** may receive and store the computer readable instructions corresponding to and/or associated with the vehicle control module **560** read from the non-volatile memory device **552**.
[0047] In at least one example embodiment, when the central memory **520** is a non-volatile memory device, the vehicle control module **560** may be stored in the central memory **520** in the form of firmware, but is not limited thereto.
[0048] The communication interface **530** may communicate data with the fourth zonal controller

400 through at least one communication bus and/or network, such as an Ethernet network, etc., but the example embodiments are not limited thereto and other communication interfaces and/or protocols may be implemented. In at least one example embodiment, the communication interface **530** may include a first physical layer (PHY) interface **531** configured to transmit and/or receive data over the communication bus (and/or a first PHY of the communication bus, etc.) to one or more other components of the vehicle, etc.

[0049] In at least one example embodiment, the first physical layer interface **531** may receive the target speed information from the central processor **510**, and may provide the target speed information to the fourth zonal controller **400**. In at least one example embodiment, the first physical layer interface **531** may receive information on a rotation speed of the motor from the fourth zonal controller **400**, and may provide the information on the rotation speed of the motor to the central processor **510**, etc.

[0050] The sensor interface **540** may receive the travel speed information, the speed change information, and/or the gear information from the sensor **600**, and may provide the travel speed information, the speed change information, and/or gear information to the central processor **510**, etc.

[0051] The storage device **550** may operate under the control of the central processor **510**, but is not limited thereto. The storage device **550** may include a storage controller **551** and the non-volatile memory device **552**, etc.

[0052] The non-volatile memory device **552** may store data. The non-volatile memory device **552** may operate in response to the control of the storage controller **551**, etc. In at least one example embodiment, the non-volatile memory device **552** may be a NAND flash memory, but is not limited thereto.

[0053] The non-volatile memory device **552** may receive at least one command and/or address from the storage controller **551**, and may perform a memory operation indicated by the command with respect to a region selected by and/or indicated by the corresponding address. The non-volatile memory device **552** may perform a program operation (e.g., write operation, etc.) for storing data in a region selected by the address, a read operation for reading data, and/or an erase operation for erasing data, etc.

[0054] The storage controller **551** may control overall operation of the storage device **550**.

[0055] In at least one example embodiment, the fourth zonal controller **400** may control operations performed in the fourth zone Z**4** among the first to fourth zones Z**1** to Z**4** of the vehicle **10**, but is not limited thereto, and for example, may control operations performed in two or more zones, etc. The fourth zonal controller **400** may control the motor **405** configured to drive the vehicle **10**, but the example embodiments are not limited thereto.

[0056] In at least one example embodiment, the fourth zonal controller **400** may include a communication interface **410**, a zonal processor **420**, a zonal memory **430**, a pulse-width modulation (PWM) interface **440**, first analog-to-digital converter (ADC) **450**, and/or a second ADC **460**, etc. According to some example embodiments, one or more of the communication interface **410**, the zonal processor **420**, the zonal memory **430**, the PWM interface **440**, the first ADC **450**, and/or the second ADC **460**, etc., may be implemented as processing circuitry. Processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0057] The communication interface **410** may communicate data with the central controller **500** through at least one communication bus, such as an Ethernet bus and/or network, but the example embodiments are not limited thereto, and for example, other communication protocols and/or interfaces may be implemented. In at least one example embodiment, the communication interface **410** may include an interface for a second physical layer interface **411** (e.g., second PHY interface, etc.) configured to transmit and/or receive data over a second PHY of the communication bus to one or more other components of the vehicle, etc. According to some example embodiments, the first PHY and the second PHY may be separate physical layers of the communication bus, or may be combined and/or the same physical layer of the communication bus, but the example embodiments are not limited thereto.

[0058] In at least one example embodiment, the second physical layer interface **411** may receive the target speed information from the central controller **500**, and may provide the target speed information to the zonal processor **420**, etc. In at least one example embodiment, the second physical layer interface **411** may receive the information on the rotation speed of the motor from the zonal processor **420**, and may provide the information on the rotation speed of the motor to the central controller **500**.

[0059] The zonal processor **420** may control overall operation of the fourth zonal controller **400**, but is not limited thereto. In at least one example embodiment, the zonal processor **420** may execute a motor control module **470** stored in the zonal memory **430**. In at least one example embodiment, the zonal processor **420** may adjust the driving voltage applied to the motor **405** based on the target speed information received from the central controller **500** and/or information on the driving current corresponding to the driving voltage applied to the motor **405**, etc.

[0060] The zonal memory **430** may be used as a buffer memory, a cache memory, an operating memory, or the like, of the fourth zonal controller **400**. In at least one example embodiment, the zonal memory **430** may store the motor control module **470**. In at least one example embodiment, the zonal memory **430** may be a volatile memory device and/or non-volatile memory device. For example, the zonal memory **430** may be a random-access memory (RAM) and/or read-only memory (ROM).

[0061] In at least one example embodiment, when a zonal memory **430** is a volatile memory device, computer readable instructions corresponding to and/or associated with the motor control module **470** may be stored in the non-volatile memory device **552** of the storage device **550**, and the zonal memory **430** may receive and/or store the motor control module **470** from the non-volatile memory device **552**, etc.

[0062] In at least one example embodiment, when the zonal memory **430** is a non-volatile memory device, computer readable instructions corresponding to and/or associated with the motor control module **470** may be stored in the zonal memory **430**, in the form of firmware, etc.

[0063] A PWM interface **440** may generate the PWM signal for controlling the width of the driving voltage to be applied to and/or used to control the motor, and may provide the PWM signal to the inverter **404**. Switches included in the inverter **404** may be turned-on and/or turned-off according to the PWM signal. The PWM interface **440** may generate the PWM signal according to the control of the zonal processor **420**.

[0064] A first ADC **450** may receive the information on the driving current corresponding to the driving voltage to be applied to the motor **405**. The first ADC **450** may convert the information on the driving current from an analog signal to a digital signal. The first ADC **450** may provide the information on the driving current to the zonal processor **420**, but is not limited thereto.

[0065] The second ADC **460** may receive the information on a rotor position (e.g., an angle and/or displacement speed, etc.) of the motor from a resolver **460**. The second ADC **460** may convert the information on the rotor position of the motor from an analog signal to a digital signal. The second ADC **460** may provide the information on the rotor position of the motor to the zonal processor **420**, but is not limited thereto.

[0066] FIG. **3** is a block diagram of an example central controller according to at least one example embodiment.

[0067] Referring to FIG. **3**, the electronic system **50** may include the fourth zonal controller **400**, the central controller **500**, and/or the sensor **600**, etc., but the example embodiments are not limited thereto. The fourth zonal controller **400** may be a controller controlling the motor **405**, but is not limited thereto.
[0068] The sensor **600** may include at least one of a speed sensor **610** (e.g., a speedometer, etc.), an accelerator pedal sensor **620**, a brake pedal sensor **630**, and/or the transmission sensor **640** (e.g., a gear selection sensor), or any combinations thereof. Additionally, in some example embodiments, the sensor **600** may further include a clutch pedal sensor, an odometer, a location sensor (e.g., GSP sensor, etc.), a compass, an altimeter, etc. Further, in some example embodiments, the sensor **600** may be safety, ADAS, and/or autonomous driving related sensors, such as radar sensors, LIDAR sensors, ultrasonic sensors, cameras (e.g., optical cameras and/or depth sensing cameras, etc.), an accelerometer, etc. The speed sensor **610** may detect the current speed of the traveling vehicle. The speed sensor **610** may generate a travel speed information RUN_INFO representing the current speed of the traveling vehicle. The speed sensor **610** may provide the travel speed information RUN_INFO to the central controller **500**, but is not limited thereto.
[0069] The accelerator pedal sensor **620** may detect a level of depressing the accelerator pedal by a driver, or in other words, the accelerator pedal sensor **620** may sense and/or detect a level and/or amount of desired acceleration instructed by the driver via the accelerator pedal, etc. The accelerator pedal sensor **620** may generate an acceleration information ACCEL_INFO representing a desired change in speed to which the vehicle **10** will increase by based on the level of depressing the accelerator pedal by a driver. However, the example embodiments are not limited thereto, and for example for a vehicle with a one-pedal driving mode, the acceleration information ACCEL_INFO may further include deceleration and/or braking information based on the level of depressing the accelerator pedal by the driver, etc. The accelerator pedal sensor **620** may provide the acceleration information ACCEL_INFO to the central controller **500**, but is not limited thereto.
[0070] The brake pedal sensor **630** may detect a level of depressing the brake pedal by the driver, or in other words, the brake pedal sensor **630** may sense and/or detect a desired level and/or amount of braking to be applied to the vehicle via the accelerator pedal, etc. The brake pedal sensor **630** may generate a deceleration information DEACCEL_INFO representing a desired change in speed to which the vehicle **10** will decrease based on the level of depressing the brake pedal by the driver. The brake pedal sensor **630** may provide the deceleration information DEACCEL_INFO to the central controller **500**, but is not limited thereto.
[0071] The transmission sensor **640** may detect the position of the transmission (e.g., detect whether the vehicle is in Park, Drive, Reverse, Neutral, first gear, second gear, third gear, etc.), but is not limited thereto. For example, the transmission sensor **640** may detect other driving related settings of the vehicle, such as a 4×4 mode, an all-wheel drive mode, a traction control mode, a comfort mode, a sports mode, a fuel/energy savings mode, a hybrid mode, etc. The transmission sensor **640** may generate a gear information GEAR_INFO representing the position of the transmission and/or other drive related settings of the vehicle. The transmission sensor **640** may provide the gear information GEAR_INFO to the central controller **500**, but is not limited thereto. According to some example embodiments, the
[0072] The central controller **500** may include the central processor **510**, the communication interface **530**, and/or the sensor interface **540**, etc., but is not limited thereto.
[0073] The sensor interface **540** may receive vehicle operation related information, such as the travel speed information RUN_INFO, the acceleration information ACCEL_INFO, the deceleration information DEACCEL_INFO, and/or the gear information GEAR_INFO, etc., from the sensor **600**, but is not limited thereto, and for example, the sensor interface **540** may receive safety related information, such as collision warning information (not shown), etc., steering warning information (not shown) indicating that the vehicle **10** has deviated from its current traffic lane, autonomous driving information (not shown) indicating autonomous driving related information (such as

steering direction information, navigation information, etc.), from the sensor **600**, but the example embodiments are not limited thereto. The sensor interface **540** may provide the travel speed information RUN_INFO, the acceleration information ACCEL_INFO, the deceleration information DEACCEL_INFO, and/or the gear information GEAR_INFO, etc., to the central processor **510**.

[0074] The communication interface **530** may include the first physical layer interface (e.g., a first PHY layer interface, etc.) **531**. The first physical layer interface **531** may receive the information on the rotation speed of the motor SPEED_ROT from the fourth zonal controller **400** over the communication bus, e.g., the first PHY of the communication bus, but is not limited thereto. The first physical layer interface **531** may provide a rotation speed information SPEED_ROT of the motor to the central processor **510**, but is not limited thereto.

[0075] In at least one example embodiment, the central processor **510** may execute the vehicle control module **560**. In at least one example embodiment, the vehicle control module **560** may determine the target speed corresponding to the speed at which the vehicle will travel based on the travel speed information RUN_INFO, the acceleration information ACCEL_INFO, the deceleration information DEACCEL_INFO, the gear information GEAR_INFO, and/or the rotation speed information SPEED_ROT of the motor, etc., and may generate a target speed information SPEED_TAR representing the target speed, but is not limited thereto, and for example, may also determine target steering commands corresponding to the direction the vehicle will travel based on current steering information, current location information, navigation information (e.g., desired destination information), etc. The central processor **510** may provide the target speed information SPEED_TAR to the fourth zonal controller **400** through the first physical layer interface **531**, etc.

[0076] FIG. **4** is a block diagram of an example zonal controller according to at least one example embodiment.

[0077] Referring to FIG. **4**, the electronic system **50** may include the fourth zonal controller **400**, the battery **403**, the inverter **404**, the motor **405**, and/or the resolver **406**, etc. While the at least one example embodiment of FIG. **4** is directed towards the control of an electric motor, the example embodiments, and for example, an internal combustion engine may be controlled as well, etc.

[0078] The fourth zonal controller **400** may include the communication interface **410**, the zonal processor **420**, the PWM interface **440**, the first ADC **450**, and/or the second ADC **460**, etc., but is not limited thereto.

[0079] The communication interface **410** may include the second physical layer interface **411** over the communication bus (e.g., over a second PHY layer of the communication bus, etc.), but is not limited thereto. The second physical layer interface **411** may receive the target speed information SPEED_TAR from the central controller **500**. The second physical layer interface **411** may provide the target speed information SPEED_TAR to the zonal processor **420**, but is not limited thereto.

[0080] The first ADC **450** may receive the information on the driving current DRIVING CURRENT corresponding to the driving voltage applied to the motor **405**, but is not limited thereto. The first ADC **450** may convert the information on the driving current DRIVING CURRENT from an analog signal to a digital signal, and provide it to the zonal processor **420**.

[0081] The second ADC **460** may receive a rotor position information POSITION of the motor from the resolver **406**, but is not limited thereto. The second ADC **460** may provide the rotor position information POSITION of the motor to the zonal processor **420**.

[0082] In at least one example embodiment, the zonal processor **420** may execute the motor control module **470**, or more specifically, the computer readable instructions associated with and/or corresponding to the motor control module, etc. When the zonal processor **420** executes the computer readable instructions of the motor control module **470**, the zonal processor **420** may generate a 3-phase voltage information 3 PHASE_VOL used for generating the PWM signal based on the target speed information SPEED_TAR, the information on the driving current DRIVING CURRENT, and/or the rotor position information POSITION of the motor, etc., but the example embodiments are not limited thereto, and for example, the voltage information may be a single-phase voltage information. Additionally, the zonal processor **420** may execute the computer readable instructions of the motor control module **470** to generate the rotation speed information SPEED_ROT of the motor based on the rotor position information POSITION of the motor, and may provide the rotation speed information SPEED_ROT of the motor to the central controller **500** through the second physical layer interface **411**, etc., but is not limited thereto.

[0083] In at least one example embodiment, the motor control module **470** may include a current controller **471**, inverse direct-quadrature (DQ) converter **472**, a DQ converter **473**, a lookup table **474**, a decoder **475**, and/or a speed calculator **476**, etc., but the example embodiments are not limited thereto, and for example, the decoder **475** may be omitted and the rotor information generated by a resolver may be used, etc. According to some example embodiments, one or more of the motor control module **470**, the current controller **471**, the inverse direct-quadrature (DQ) converter **472**, the DQ converter **473**, the lookup table **474**, the decoder **475**, and/or the speed calculator **476**, etc., may be implemented as processing circuitry. Processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0084] A decoder **475** may receive the rotor position information POSITION of the motor from the second ADC **460**, but is not limited thereto. The decoder **475** may identify a rotor angle of the motor based on the rotor position information POSITION of motor. The decoder **475** may generate rotor angular information ANGLE based on the rotor position information POSITION of motor, etc. The decoder **475** may provide the rotor angular information ANGLE to the DQ converter **473** and/or the speed calculator **476**, etc.

[0085] The speed calculator **476** may calculate the rotation speed of the motor based on the rotor angular information ANGLE. The speed calculator **476** may generate the rotation speed information SPEED_ROT of the motor based on the rotor angular information ANGLE. The speed calculator **476** may provide the rotation speed information SPEED_ROT of the motor to the central controller **500** through the communication interface **410**, etc.

[0086] The DQ converter **473** may receive the information on the driving current DRIVING CURRENT from the first ADC **450**. In at least one example embodiment, the information on the driving current DRIVING CURRENT may be information representing a value of the driving current flowing between the inverter **404** and the motor **405** as the driving voltage is applied to the motor **405**. In at least one example embodiment, the driving current may be a 3-phase current, but is not limited thereto, and for example, may be a single-phase current, etc.

[0087] In at least one example embodiment, the DQ converter **473** may receive the rotor angular information ANGLE from the decoder **475**, and may receive a sine-cosine angular information SIN-COS ANGLE representing sine value and cosine value corresponding to the rotor angle from the lookup table **474**. The DQ converter **473** may provide the sine-cosine angular information SIN-COS ANGLE to an inverse DQ converter **472**.

[0088] In at least one example embodiment, the DQ converter **473** may perform a DQ conversion that converts the driving current to a DQ-axis driving current based on the information on the driving current DRIVING CURRENT and the sine-cosine angular information SIN-COS ANGLE.

[0089] In at least one example embodiment, the DQ conversion may include a Clarke conversion that converts the value of the driving current to a value of an alpha-beta axis current of a stationary reference frame and a Park conversion that converts the value of the alpha-beta axis current to a value of a DQ-axis current of a rotor reference frame, but the example embodiments are not limited thereto.

[0090] In at least one example embodiment, the DQ converter **473** may convert the value of the

driving current to the DQ-axis the value of the driving current, and may generate a DQ-axis driving current information DQ_DRIVING CURRENT representing the DQ-axis the value of the driving current. The DQ converter **473** may provide the DQ-axis driving current information DQ_DRIVING CURRENT to the current controller **471**.
[0091] The current controller **471** may receive the target speed information SPEED_TAR and the DQ-axis driving current information DQ_DRIVING CURRENT. In at least one example embodiment, the target speed information SPEED_TAR may be information representing the target speed at which the vehicle **10** will travel as a value of a target current. The current controller **471** may determine a DQ-axis target voltage based on the result of comparison between the DQ-axis driving current and the target current corresponding to the target speed. In at least one example embodiment, the current controller **471** may include Proportional-Integral Controller.
[0092] The current controller **471** may generate DQ-axis target voltage information DQ_TAR_VOL representing the value of the DQ-axis target voltage based on the target speed information SPEED_TAR and the DQ-axis driving current information DQ_DRIVING CURRENT. The current controller **471** may provide the DQ-axis target voltage information DQ_TAR_VOL to the inverse DQ converter **472**.
[0093] In at least one example embodiment, the inverse DQ converter **472** may perform an inverse DQ conversion that converts the DQ-axis target voltage to a 3-phase voltage based on the DQ-axis target voltage information DQ_TAR_VOL and the sine-cosine angular information SIN-COS ANGLE. In at least one example embodiment, the inverse DQ conversion may be an operation to convert the value of the DQ-axis target voltage to a value of 3-phase voltage.
[0094] In at least one example embodiment, the inverse DQ converter **472** may convert the value of the DQ-axis target voltage to a value of 3-phase voltage, and may generate the 3-phase voltage information 3 PHASE_VOL representing the value of the 3-phase voltage. The inverse DQ converter **472** may provide the 3-phase voltage information 3 PHASE_VOL to the PWM interface **440**.
[0095] The PWM interface **440** may generate a PWM signal PWM_SIG based on the 3-phase voltage information 3 PHASE_VOL received from the inverse DQ converter **472**. The PWM signal PWM_SIG may be a signal to control the width of the driving voltage applied to the motor **405**. The PWM interface **440** may provide the PWM signal PWM_SIG to the inverter **404**. Switches included in the inverter **404** may be turned-on and/or turned-off according to the PWM signal PWM_SIG.
[0096] According to the electronic system **50** included in the vehicle **10** according to at least one of the example embodiments of the inventive concepts, a central controller controlling a plurality of zonal controllers may determine the target speed corresponding to the desired speed at which the vehicle **10** will travel, and one zonal controller controlling the operation performed in one zone among a plurality of zones may control at least one motor configured to drive the vehicle **10**, thereby improving the performance of controlling the vehicle in the longitudinal direction.
[0097] FIG. **5** is a diagram of an example inverter, motor, and zonal controller according to at least one example embodiment.
[0098] Referring to FIG. **5**, the electronic system **50** may include the fourth zonal controller **400**, the battery **403**, the inverter **404**, the motor **405**, and/or the resolver **406**, etc., but is not limited thereto.
[0099] The fourth zonal controller **400** may include the zonal processor **420**, the PWM interface **440**, the first ADC **450**, and/or the second ADC **460**, etc., but is not limited thereto.
[0100] The battery **403** may include a power DC (e.g., DC power supply, DC power source, etc.) and/or a capacitor CP, etc. The battery **403** may be connected to an inverter **405**. The power DC may supply a DC power. The capacitor CP may store electrical energy. The capacitor CP may output DC power supplied by the power DC.
[0101] In at least one example embodiment, the zonal processor **420** may generate the 3-phase

voltage information 3 PHASE_VOL based on the target speed information received from the central controller **500**, the information on the driving current DRIVING CURRENT received through the first ADC **450**, and/or the rotor position information POSITION of the motor received through the second ADC **460**, etc., but is not limited thereto. The zonal processor **420** may provide the 3-phase voltage information 3 PHASE_VOL to the PWM interface **440**.

[0102] The PWM interface **440** may generate a PWM signal based on the 3-phase voltage information 3 PHASE_VOL. The PWM signal may include a plurality of signals, e.g., first to sixth signals SIG**1** to SIG**6**, but is not limited thereto, and for example, may include a greater or lesser number of signals, etc. The PWM interface **440** may apply the first to sixth signals SIG**1** to SIG**6** to the inverter **404** based on the 3-phase voltage information 3 PHASE_VOL, to for example, instruct and/or control the inverter **404** to turn-on and/or turn-off first to sixth switches T**1** to T**6** included in the inverter **404**, etc.

[0103] The inverter **404** may apply the driving voltage converted from the DC power supplied from the battery **403** to the motor **405**. In at least one example embodiment, the driving voltage may be a 3-phase voltage, but is not limited thereto, and for example, may be a single-phase voltage, etc. 3-phase voltage may include an A-phase voltage Va, a B-phase voltage Vb, and a C-phase voltage Vc.

[0104] In at least one example embodiment, the inverter **404** may include a first switch T**1**, a second switch T**2**, a third switch T**3**, a fourth switch T**4**, a fifth switch T**5**, and/or a sixth switch T**6**, etc., but is not limited thereto.

[0105] In at least one example embodiment, the first switch T**1** and the second switch T**2** may be turned-on and/or turned-off according to and/or based on a first signal SIG**1** and/or a second signal SIG**2** received from the PWM interface **440**, etc. The width of the A-phase voltage Va applied to the motor **405** may be changed according to and/or corresponding to the time at which the first switch T**1** and/or the second switch T**2** are to be turned-on and/or turned-off. The time at which the first switch T**1** and/or the second switch T**2** are turned-on and/or turned-off may be changed according to the first signal SIG**1** and/or the second signal SIG**2**, etc. In other words, the zonal processor **420** may control the first switch T**1** and/or second switch T**2** using at least one control signal (e.g., PWM signal, etc.).

[0106] In at least one example embodiment, the third switch T**3** and/or the fourth switch T**4** may be turned-on and/or turned-off according to a third signal SIG**3** and/or a fourth signal SIG**4** received from the PWM interface **440**, etc. The width of a B-phase voltage Vb applied to the motor **405** may be changed according to and/or corresponding to the time at which the third switch T**3** and/or the fourth switch T**4** are to be turned-on and/or turned-off. The time at which the third switch T**3** and/or the fourth switch T**4** are turned-on and/or turned-off may be changed according to and/or based on the third signal SIG**3** and/or the fourth signal SIG**4**, etc. In other words, the zonal processor **420** may control the third switch T**3** and/or the fourth switch T**4** using at least one control signal (e.g., PWM signal, etc.).

[0107] In at least one example embodiment, the fifth switch T**5** and/or the sixth switch T**6** may be turned-on and/or turned-off according to a fifth signal SIG**5** and/or a sixth signal SIG**6** received from the PWM interface **440**. The width of the C-phase voltage Vc applied to the motor **405** may be changed according to and/or corresponding to the time at which the fifth switch T**5** and/or the sixth switch T**6** are to be turned-on and/or turned-off, etc. The time at which the fifth switch T**5** and/or the sixth switch T**6** are turned-on and/or turned-off may be changed according to and/or based on the fifth signal SIG**5** and the sixth signal SIG**6**. In other words, the zonal processor **420** may control the fifth switch T**5** and/or the sixth switch T**6** using at least one control signal (e.g., PWM signal, etc.).

[0108] In at least one example embodiment, the PWM interface **440** may adjust the time at which the plurality of switches, e.g., the first to sixth switches T**1** to T**6**, etc., are turned-on and/or turned-off based on the plurality of signals, e.g., the first to sixth signals SIG**1** to SIG**6**, etc., and the widths of the A-phase voltage Va, B-phase voltage Vb, and/or the C-phase voltage Vc applied to the motor **405** may be adjusted according to and/or corresponding to the time at which the first to sixth switches T**1** to T**6** are to be turned-on and/or turned-off, etc.

[0109] The first ADC **450** may receive the driving current flowing between the inverter **404** and the motor **405** as the A-phase voltage Va, B-phase voltage Vb, and the C-phase voltage Vc are applied to the motor **405**. In at least one example embodiment, the driving current may be a 3-phase current, but is not limited thereto, and for example, may be a single-phase current. 3-phase current may include an A-phase current Ia, a B-phase current Ib, and a C-phase current Ic. The first ADC **450** may provide the information on the driving current DRIVING CURRENT representing values of the A-phase current Ia, the B-phase current Ib, and the C-phase current Ic to the zonal processor **420**, etc.

[0110] The resolver **406** may detect the rotor position of the motor. The resolver **406** may provide the rotor position information POSITION representing the rotor position of the motor to the zonal processor **420** through the second ADC **460**, but is not limited thereto.

[0111] FIG. **6** is a block diagram of example zonal controllers for controlling a plurality of motors according to at least one example embodiment.

[0112] Referring to FIG. **6**, the vehicle **10** may include the electronic system **50**, but is not limited thereto. The vehicle **10** may include a plurality of zones, such as the first zone Z**1**, the second zone Z**2**, the third zone Z**3**, and/or the fourth zone Z**4**, etc., but is not limited thereto, and for example, may include a greater or lesser number of zones, etc. In at least one example embodiment, the first zonal controller **100** may control the first zone Z**1**, the second zonal controller **200** may control the second zone Z**2**, the third zonal controller **300** may control the third zone Z**3**, and/or the fourth zonal controller **400** may control the fourth zone Z**4**, etc.

[0113] In at least one example embodiment, the second zonal controller **200** may control at least one first motor **204**, etc. In at least one example embodiment, the first motor **204** may be at least one motor to drive at least one wheel located in the front of the vehicle (e.g., a wheel located in the first zone of the vehicle, etc.), but is not limited thereto. In at least one example embodiment, the fourth zonal controller **400** may control at least one second motor **408**. In at least one example embodiment, the second motor **408** may be at least one motor configured to drive at least one wheel located in the rear of the vehicle (e.g., a wheel located in the second zone of the vehicle, etc.), but is not limited thereto.

[0114] In at least one example embodiment, as described in connection with FIG. **2**, the fourth zonal controller **400** may include the communication interface **410**, the zonal processor **420**, the zonal memory **430**, the PWM interface **440**, the first ADC **450**, and/or the second ADC **460**, etc.

[0115] In at least one example embodiment, the second zonal controller **200** may be implemented in the same way as the fourth zonal controller **400**, but is not limited thereto. In at least one example embodiment, the second zonal controller **200** may include at least one communication interface (e.g., Ethernet interface, etc.) configured to receive the target speed information from the central controller **500**, a first ADC configured to receive first information on the driving current corresponding to a first driving voltage applied to the first motor **204**, a second ADC configured to receive rotor position information of the first motor from a first resolver **203**, a zonal processor configured to generate the 3-phase voltage information based on the target speed information, the first information on the driving current, and/or the rotor position information of the first motor, etc., a zonal memory, and/or a PWM interface configured to generate the PWM signal for turning-on and/or turning-off switches included in a first inverter **205** based on the 3-phase voltage information, etc.

[0116] In at least one example embodiment, the first inverter **205** may convert voltage of the battery **403** to the first driving voltage, and may apply the first driving voltage to the at least one first motor **204**. The at least one first motor **204** may drive at least one wheel connected to the at least one first motor **204** according to and/or based on the first driving voltage. A second inverter

409 may convert voltage of the battery **403** to a second driving voltage, and may apply the second driving voltage to the at least one second motor **408**. The at least one second motor **408** may drive at least one wheel connected to the at least one second motor **408** according to the second driving voltage. While FIG. **5** illustrates two motors in the vehicle, the example embodiments are not limited thereto, and for example, the vehicle may include a single motor or three or more motors for driving the wheels of the vehicle, etc.

[0117] In at least one example embodiment, the first resolver **203** may detect a rotor position of the first motor **204** and a second resolver **409** may detect a rotor position of the second motor **408**, but the example embodiments are not limited thereto, and there may be a greater or lesser number of resolvers and/or motors, etc.

[0118] In at least one example embodiment, the central controller **500** may receive the travel speed information representing a speed of the vehicle, the speed change information representing the speed to which the vehicle will increase or decrease, and/or the gear information representing the position of the transmission from the sensor **600**, etc. The central controller **500** may receive first rotation speed information representing a speed at which the first motor **204** rotates from the second zonal controller **200**, etc. The central controller **500** may receive second rotation speed information representing a speed at which the second motor **408** rotates from the fourth zonal controller **400**, etc.

[0119] In at least one example embodiment, the central controller **500** may determine the target speed corresponding to the speed at which the vehicle will travel based on the travel speed information, the speed change information, the gear information, first rotation speed information, and/or second rotation speed information, etc., but is not limited thereto. The central controller **500** may provide the target speed information representing the target speed to the second zonal controller **200** and/or the fourth zonal controller **400**, etc.

[0120] In at least one example embodiment, the second zonal controller **200** may generate a first PWM signal that may adjust a width of the first driving voltage based on the target speed information, information on the rotor position of the first motor, and/or the first information on the driving current corresponding to the first driving voltage applied to the first motor **204**, etc., but is not limited thereto. The second zonal controller **200** may turn-on and/or turn-off switches included in the first inverter **205** based on the first PWM signal. The first motor **204** may drive the wheel connected to the first motor **204** according to and/or based on the first driving voltage whose width is adjusted.

[0121] In at least one example embodiment, the fourth zonal controller **400** may generate a second PWM signal that may adjust a width of the second driving voltage based on the target speed information, information on the rotor position of the second motor, and/or second information on the driving current corresponding to the second driving voltage applied to the second motor **408**, etc., but is not limited thereto. The fourth zonal controller **400** may turn-on and/or turn-off switches included in the second inverter **409** based on the second PWM signal. The second motor **408** may drive the wheel connected to the second motor **408** according to and/or based on the second driving voltage whose width is adjusted.

[0122] FIG. **7** is a flowchart illustrating an example method of operating an electronic system according to at least one example embodiment.

[0123] Referring to FIG. **7**, at operation S**70**, the central controller **500** included in the electronic system **50** may receive the speed information from the sensor. The speed information may include the travel speed information representing the speed of the travel vehicle, acceleration information representing a speed to which the vehicle will increase, the speed change information representing a speed to which the vehicle will decrease, and/or the gear information representing the position of the transmission, etc., but the example embodiments are not limited thereto. In at least one example embodiment, the central controller **500** may receive the rotation speed information representing a speed and/or revolutions per minute (RPMs) at which the at least one motor rotates from the zonal

controller controlling the motor among the plurality of zonal controllers included in the electronic system **50**, etc., but is not limited thereto.
[0124] At operation S**72**, the central controller **500** may generate the target speed information representing the desired speed at which the vehicle will travel based on the speed information. The central controller **500** may provide the target speed information to the zonal controller controlling the at least one motor among the plurality of zonal controllers.
[0125] At operation S**74**, the zonal controller controlling the at least one motor may receive the driving current information and/or the position information, etc., but is not limited thereto. The driving current information may be information representing the value of the driving current flowing between the inverter and the at least one motor according to the driving voltage applied to the at least one motor. The position information may be information representing the rotor position of the at least one motor.
[0126] At operation S**76**, the zonal controller controlling the at least one motor may generate the PWM signal based on the target speed information and/or the driving current information, etc., but is not limited thereto. The PWM signal may be at least one signal used to control the at least one motor of the vehicle by changing a width of the driving voltage applied to the at least one motor that drives the vehicle.
[0127] FIG. **8** is a flowchart illustrating an example method of operating a zonal controller according to at least one example embodiment.
[0128] In at least one example embodiment, FIG. **8** may correspond to operation S**76** of FIG. **7**. Referring to FIG. **8**, at operation S**80**, the zonal controller controlling the at least one motor among the plurality of zonal controllers may generate the rotor angular information based on rotor position information of the at least one motor. In at least one example embodiment, the zonal controller controlling the motor may calculate the rotation speed of the motor based on the rotor angular information, and may provide the information on the rotation speed of the motor to the central controller **500**.
[0129] At operation S**82**, the zonal controller controlling the motor may generate the DQ-axis driving current information based on the driving current information and/or the rotor angular information, etc. In at least one example embodiment, the zonal controller controlling the at least one motor may perform the DQ conversion that converts the value of the driving current to the DQ-axis the value of the driving current, and may generate the DQ-axis driving current information representing the value of the DQ-axis driving current.
[0130] At operation S**84**, the zonal controller controlling the at least one motor may generate the DQ-axis target voltage information based on target current information and/or the DQ-axis driving current information, etc. In at least one example embodiment, the zonal controller controlling the at least one motor may determine the DQ-axis target voltage based on the result of comparison between the DQ-axis driving current and the target current corresponding to the target speed, and may generate the DQ-axis target voltage information representing the value of the DQ-axis target voltage.
[0131] At operation S**86**, the zonal controller controlling the at least one motor may generate the 3-phase voltage information based on and/or converted from the DQ-axis target voltage information. In at least one example embodiment, the zonal controller controlling the at least one motor may perform the inverse DQ conversion that converts the DQ-axis target voltage to 3-phase voltage, and may generate the 3-phase voltage information representing the value of the 3-phase voltage.
[0132] At operation S**88**, the zonal controller controlling the at least one motor may generate the PWM signal based on the 3-phase voltage information. In at least one example embodiment, the zonal controller controlling the at least one motor may provide the PWM signal to the inverter that applies the driving voltage to the at least one motor, etc., but is not limited thereto.
[0133] While various example embodiments of the inventive concepts have been described in connection with several illustrations, it is to be understood that the example embodiments of the

inventive concepts are not limited thereto. On the contrary, it is intended that the inventive concepts cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.

Claims

1. An electronic system, comprising: a central controller configured to, receive speed information related to a current speed of a vehicle, generate a target speed information representing a speed at which the vehicle will travel based on the speed information, and transmit the target speed information; and a first zonal controller configured to control a first zone among a plurality of zones of the vehicle; and a second zonal controller configured to control a second zone among the plurality of zones of the vehicle, and control at least one motor of the vehicle based on the target speed information received from the central controller, the at least one motor configured to drive at least one wheel of the vehicle.
2. The electronic system of claim 1, wherein the second zonal controller is further configured to: receive the target speed information through a physical layer of a communication bus.
3. The electronic system of claim 1, wherein the speed information comprises: speed change information received from a pedal sensor; and rotation speed information corresponding to the at least one motor.
4. The electronic system of claim 1, wherein the second zonal controller is further configured to: control the at least one motor based on the target speed information using a pulse width modulation (PWM) signal, the controlling the at least one motor including, an inverter connected to the at least one motor, the inverter configured to apply a driving voltage to the at least one motor based on the PWM signal.
5. The electronic system of claim 1, wherein the second zonal controller comprises: a first analog-to-digital converter (ADC) configured to receive information on a driving current corresponding to a driving voltage applied to the at least one motor; and a second ADC configured to receive rotor position information of a rotor included in the at least one motor.
6. The electronic system of claim 5, wherein the second zonal controller is further configured to: generate rotation speed information corresponding to the at least one motor based on the rotor position information; and provide the rotation speed information to the central controller through a communication bus.
7. The electronic system of claim 6, wherein the second zonal controller is further configured to: generate angular information corresponding to the rotor based on the rotor position information; and generate direct-quadrature (DQ)-axis driving current information based on the driving current information and the angular information of the rotor.
8. The electronic system of claim 7, wherein the second zonal controller is further configured to: generate a DQ-axis target voltage information based on the DQ-axis driving current information and target current information corresponding to the target speed information; and generate 3-phase voltage information based on the DQ-axis target voltage information.
9. The electronic system of claim 8, wherein the second zonal controller is further configured to: adjust a width of a driving voltage to be applied to the at least one motor based on the 3-phase voltage information.
10. An electronic system, comprising: an inverter configured to convert a voltage of a battery to a driving voltage; at least one motor configured to operate based on the driving voltage; a plurality of zonal controllers configured to control each of a plurality of zones of a vehicle; and a central controller configured to determine target speed information based on current rotation speed information of the at least one motor and desired speed information of the vehicle housing the at least one motor, the target speed information indicating a target speed at which the vehicle will travel, and the desired speed information indicating a desired speed to which the vehicle will

increase or decrease, wherein one zonal controller for controlling one of the plurality of zones of the vehicle is configured to, receive the target speed, and adjust the driving voltage based on a target current corresponding to the target speed information and a driving current corresponding to the driving voltage.

11. The electronic system of claim 10, wherein the central controller is further configured to: receive the desired speed information from a sensor configured to detect a position of a pedal; and receive rotation speed information of the at least one motor from the zonal controller.

12. The electronic system of claim 10, wherein the one zonal controller is further configured to: receive the target speed information from the central controller.

13. The electronic system of claim 10, wherein the one zonal controller comprises: a first analog-to-digital converter (ADC) configured to receive information regarding the driving current; and a second ADC configured to receive information regarding a rotor position of the at least one motor.

14. The electronic system of claim 13, wherein the one zonal controller is further configured to: identify a rotor angle of the at least one motor based on information on the rotor position of the at least one motor; and convert the driving current to a direct-quadrature (DQ)-axis driving current based on the identified rotor angle.

15. The electronic system of claim 14, wherein the one zonal controller is further configured to: determine rotation speed information of the at least one motor based on the identified rotor angle; and provide the rotation speed information of the at least one motor to the central controller.

16. The electronic system of claim 14, wherein the one zonal controller is further configured to: determine a DQ-axis target voltage based on the target current and the DQ-axis driving current; and convert the DQ-axis target voltage to 3-phase voltage based on the identified rotor angle.

17. The electronic system of claim 16, wherein the one zonal controller is further configured to: control a width of the driving voltage based on information on the 3-phase voltage.

18. A vehicle, comprising: a first motor and a second motor each configured to drive at least one first wheel and at least one second wheel, respectively; a central controller configured to generate target speed information and speed change information of the vehicle, the target speed information representing a speed at which the vehicle will travel based on rotation speed information of the first motor and the second motor; a first zonal controller configured to control a first zone among a plurality of zones of the vehicle, and control a first driving voltage applied to the first motor based on the target speed information; and a second zonal controller configured to control a second zone among a plurality of zones of the vehicle, and control a second driving voltage applied to the second motor based on the target speed information.

19. The vehicle of claim 18, the vehicle further comprises: a communication bus; and wherein the central controller is further configured to receive the rotation speed information through the communication bus; and the first zonal controller and the second zonal controller are each configured to receive the target speed information through the communication bus.

20. The vehicle of claim 18, wherein the central controller is further configured to determine a target current corresponding to the target speed information based on the rotation speed information of the first motor and the second motor and the speed change information of the vehicle; and the first zonal controller and the second zonal controller each further configured to, receive rotor position information of the first motor and the second motor, respectively, and generate the rotation speed information based on the respective rotor position information.